nRF9161-DEV-BOARD | nRF9161-LACA-R7 | Hierarchical schematic

A. nRF9161

File: schematic/01_nRF9161.kicad_sch

B. POWER

File: schematic/02_POWER.kicad_sch

C. LTE/NR+ RF

File: schematic/03_LTE_RF.kicad_sch

D. GNSS

File: schematic/04_GNSS.kicad_sch

E. SIM/UICC

File: schematic/05_SIM.kicad_sch

F. SWD/DEBUG

File: schematic/06_SWD.kicad_sch

G. UART

File: schematic/07_UART.kicad_sch

H. SPI/I2C

File: schematic/08_SPI_I2C.kicad_sch

I. I2S/PDM/ADC

File: schematic/09_I2S_PDM_ADC.kicad_sch

J. GPIO EXPANSION

File: schematic/10_GPIO_EXPANSION.kicad_sch

K. LEDs/BUTTONS

File: schematic/11_LEDS_BUTTONS.kicad_sch

L. TEST POINTS

File: schematic/12_TESTPOINTS.kicad_sch

Electrical source of truth: nRF9161 Product Specification v1.0 Table 55 + Figure 122.
nRF91 Hardware Design Guidelines decoupling, nRF9161 DK 1.0.0 RF/GNSS/SIM reference.
Do not populate C18–C21 RF match until antenna/VNA characterization (Nordic OR pi default).

Sheet: /
File: nRF9161-DEV-BOARD.kicad_sch

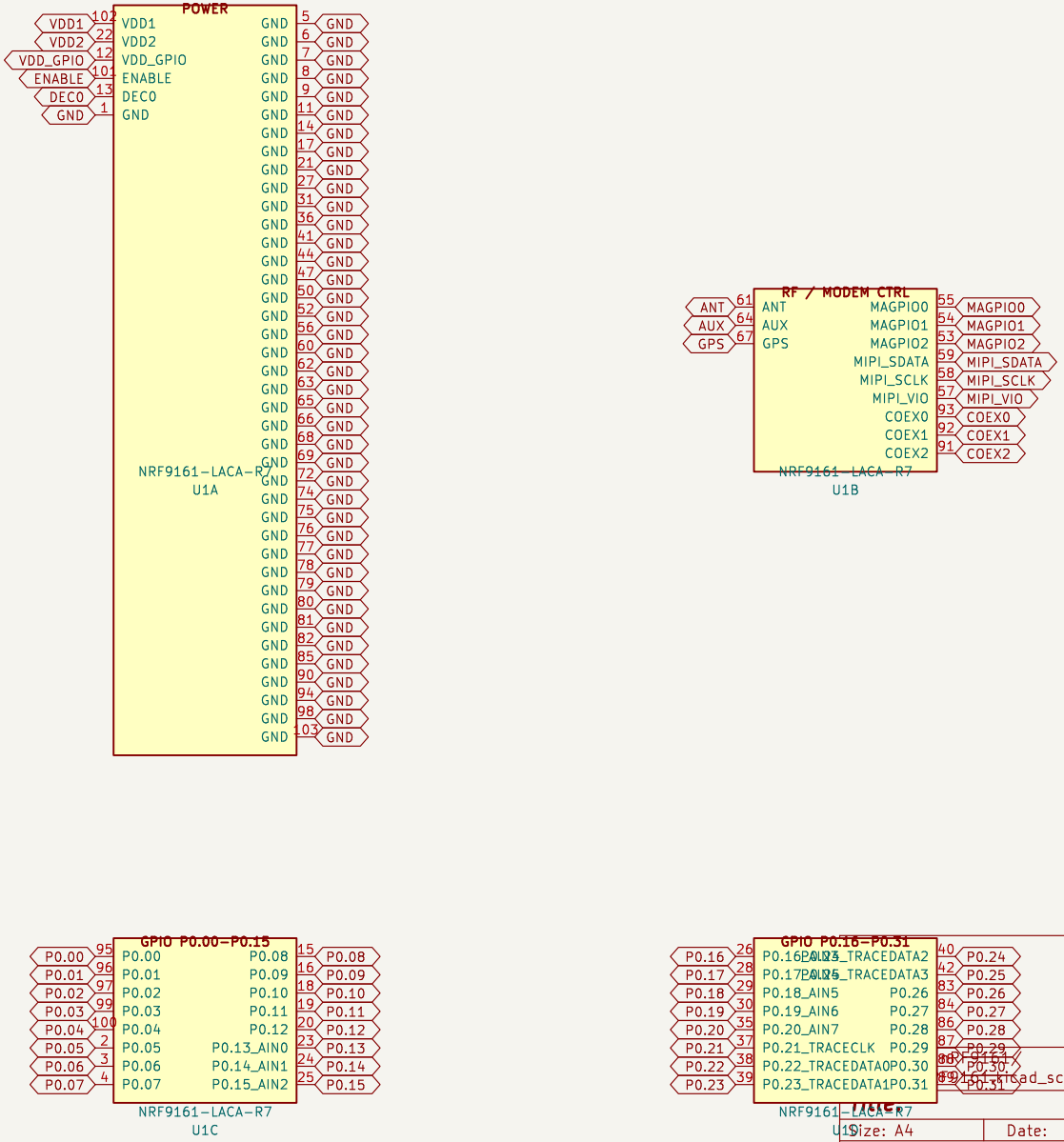
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 1/13

A. nRF9161-LACA-R7 (Nordic PS v1.0 Table 55)

SIM_DET (45) MUST FLOAT per PS. RES pins soldered mechanically, electrically NC. GND_Shield tied to VSS pin 103.

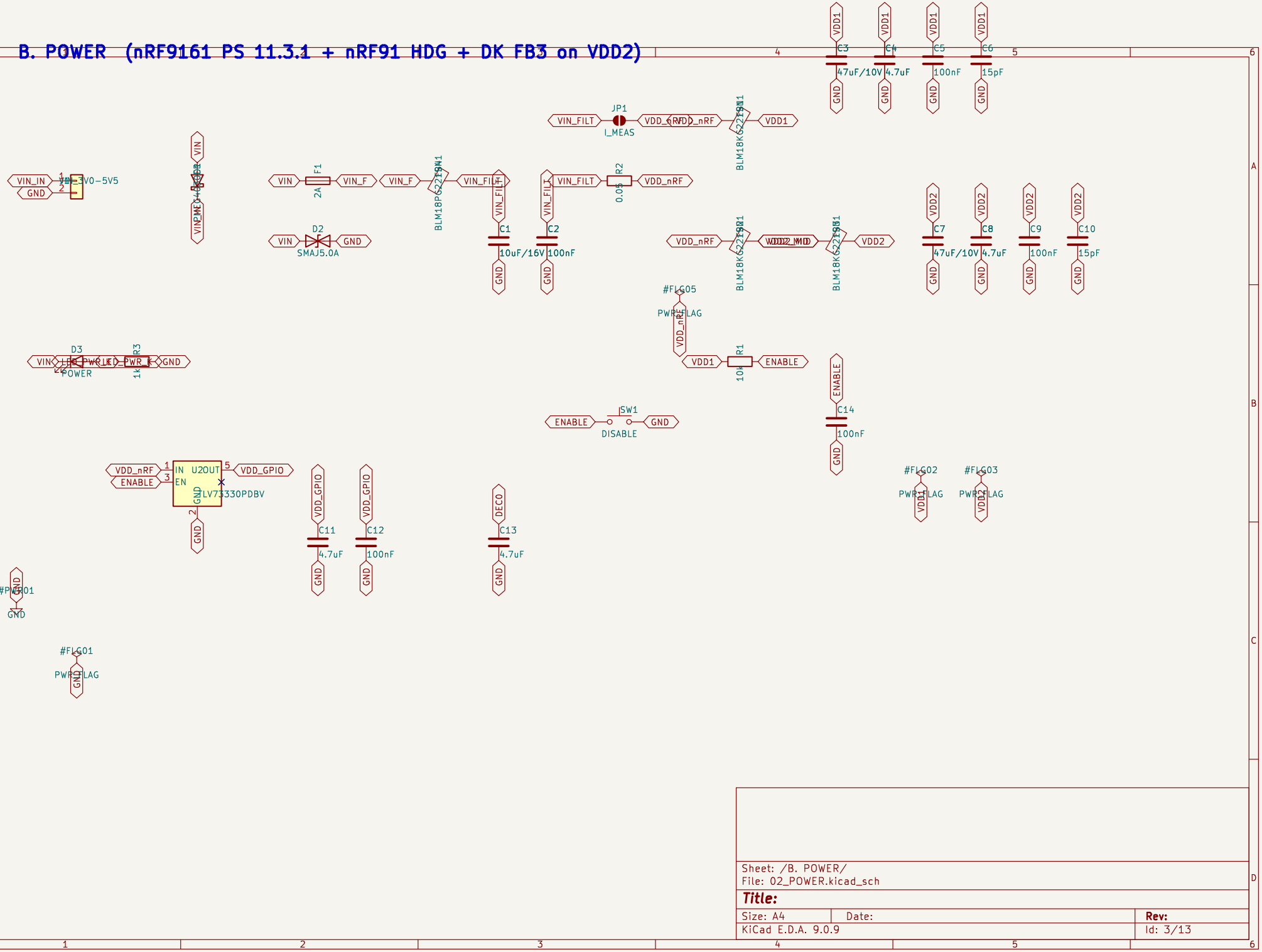
nRF9161 PIN POLICY (PS v1.0)
RES pins 10,51,70,71,73,104-127: solder for mechanical rigidity, electrically NC
SIM_DET pin 45: MUST FLOAT - do not wire (PS: Not used)
GND_Shield / VSS: board GND plane
RES pins NC electrically. SIM_DET pin 45 must float (PS). DEC0=4.7uF.
DEC0 pin 13: 4.7uF only (PCN134 / nRF91x1)



nRF9161/nRF9161.kicad_sch

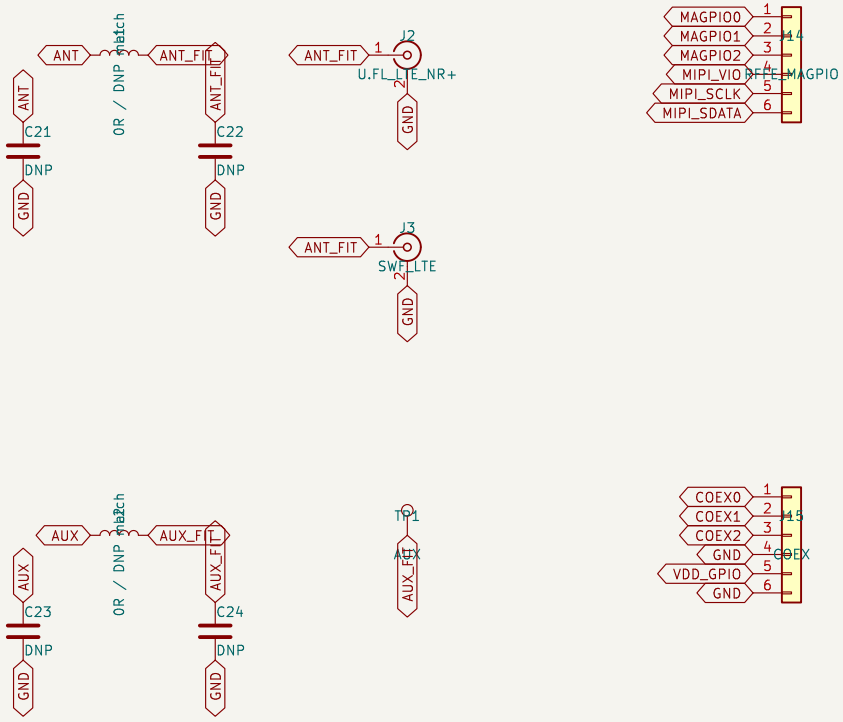
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.9		Id: 2/13

B. POWER (nRF9161 PS 11.3.1 + nRF91 HDG + DK FB3 on VDD2)



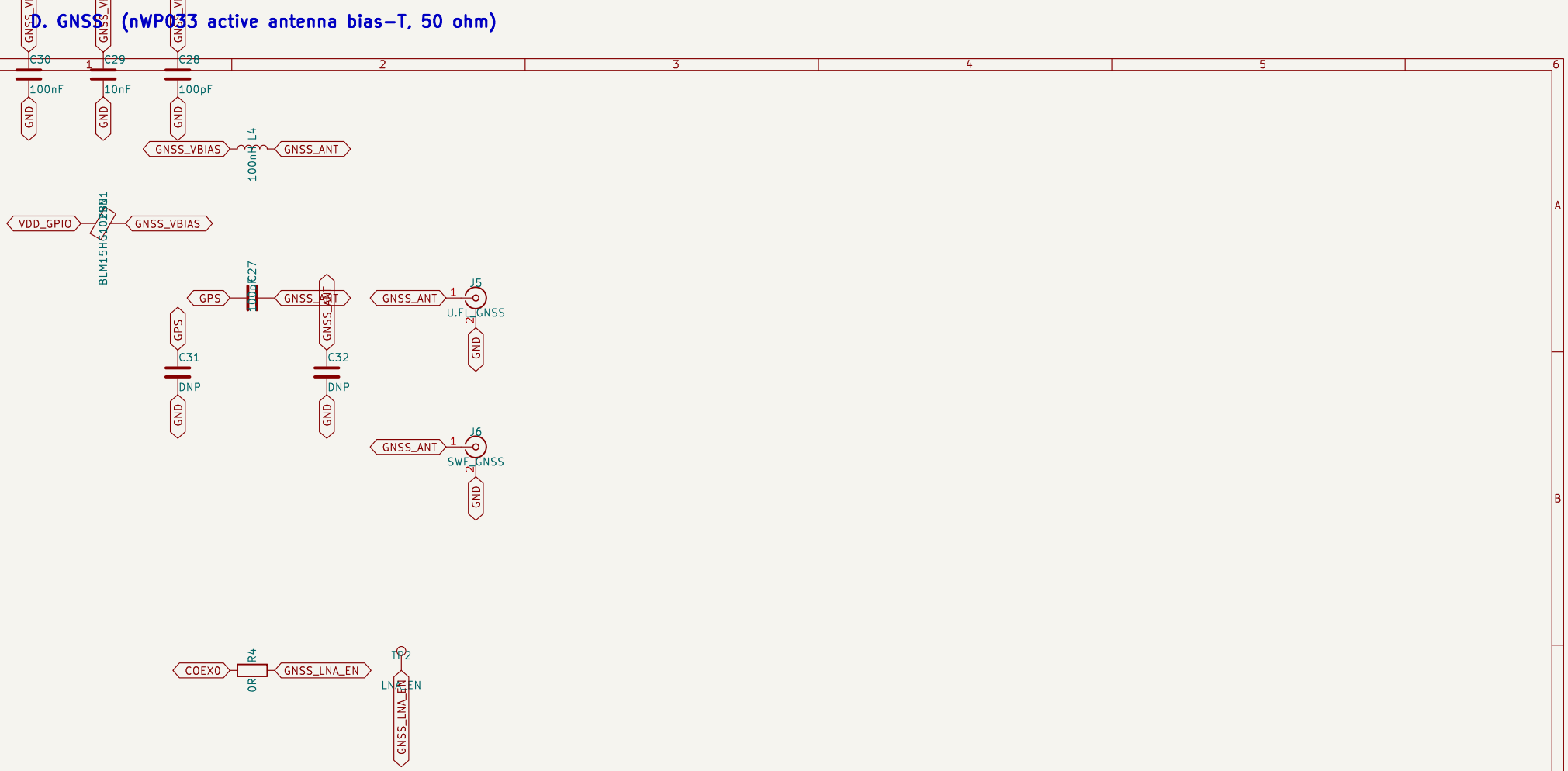
Sheet: /B. POWER/	
File: 02_POWER.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 9.0.9	**Rev:** Id: 3/13

C. LTE / DECT NR+ RF (50 ohm ANT/AUX, Nordic pi-match reserved)



Sheet: /C. LTE/NR+ RF/		
File: 03_LTE_RF.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 4/13

D. GNSS (nWP033 active antenna bias-T, 50 ohm)

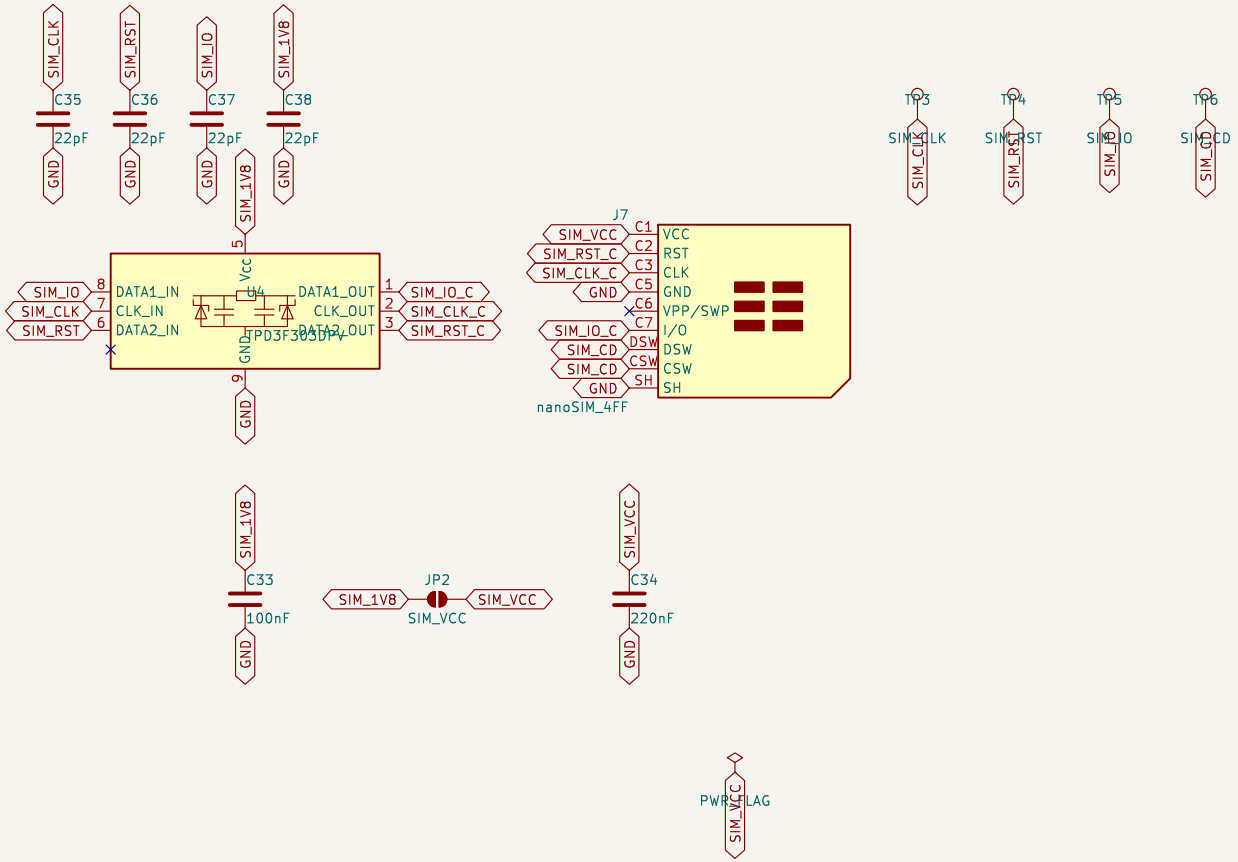


Sheet: /D. GNSS/
File: 04_GNSS.kicad_sch

Title:

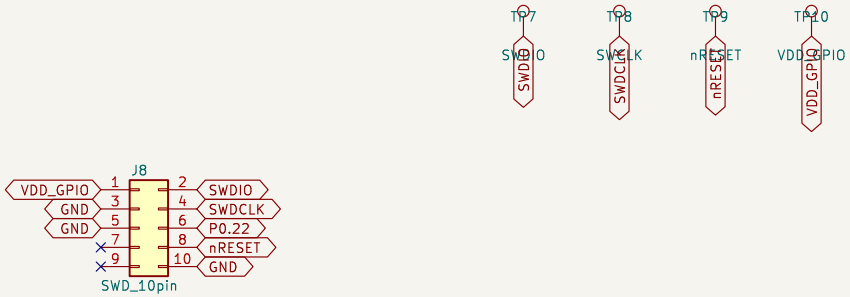
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 5/13

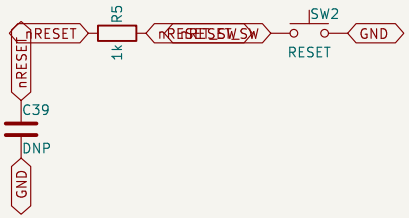
E. SIM/UICC (TPD3F303 per nRF91 HDG; SIM_DET floats)



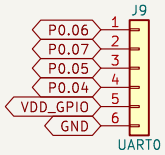
Sheet: /E. SIM/UICC/		
File: 05_SIM.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 6/13

F. SWD/DEBUG (10-pin Cortex-M, J-Link compatible)





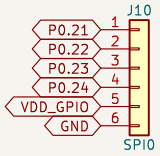
Sheet: /F. SWD/DEBUG/		
File: 06_SWD.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 7/13



Sheet: /G. UART/
File: 07_UART.kicad_sch

Title:

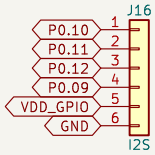
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 8/13

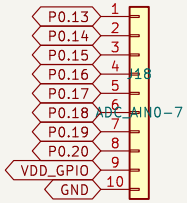


Sheet: /H. SPI/I2C/
File: 08_SPI_I2C.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 9/13





Sheet: /I. I2S/PDM/ADC/
File: 09_I2S_PDM_ADC.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 10/13

P0.22 pin38 SPI_MOSI / SWO J8/J10
P0.23 pin39 SPI_MISO J10
P0.24 pin40 SPI_CS J10
P0.25 pin42 J13 TRACEDATA3
P0.26 pin83 J13
P0.27 pin84 J13
P0.28 pin86 J13
P0.29 pin87 J13
P0.30 pin88 I2C SDA J11
P0.31 pin89 I2C SCL J11
Headers are labels only except LEDs and USER.
Do not wire SIM_DET (pin 45); it must float.

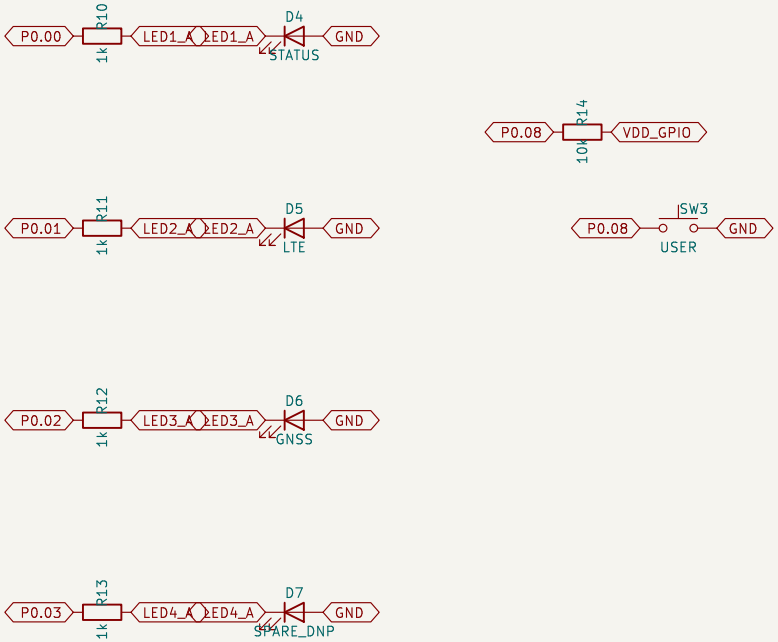
GPIO / INTERFACE MAP (nRF9161 PS Table 55). All 32 GPIOs on J12/J13.
P0.00–P0.03 LED1–4 (open–drain OK; still on headers)
P0.04 UART_CTS P0.05 UART_RTS P0.06 UART_TX P0.07 UART_RX (J9 default)
P0.08 USER button (SW3, 10k to VDD_GPIO)
P0.09 I2S_SDIN P0.10 I2S_SCK P0.11 I2S_LRCK P0.12 I2S_SDOUT (J16)
P0.13–P0.20 AIN0–AIN7 (J18) P0.17/P0.18 also PDM CLK/DIN (J17)
P0.21 SPI_SCK / TRACECLK P0.22 SPI_MOSI / SWO P0.23 SPI_MISO P0.24 SPI_CS (J10)
P0.25 TRACE_D3 P0.26–P0.29 expansion P0.30 SDA P0.31 SCL (J11)
Default UART/SPI/I2C/I2S/PDM are header labels only – GPIOs are not dedicated in hardware except LEDs/USER.

J12 — GPIO_P0.00-15

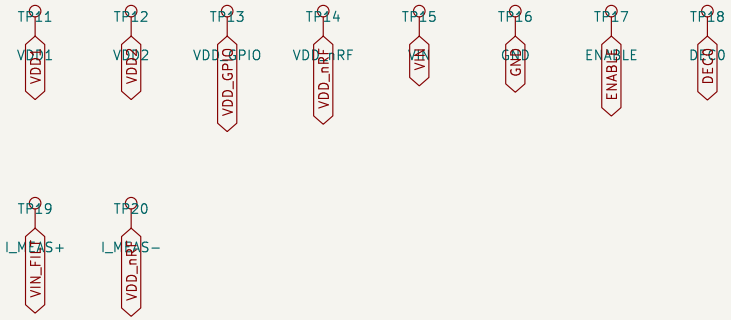
Pin	Net
1	P0.00
2	P0.01
3	P0.02
4	P0.03
5	P0.04
6	P0.05
7	P0.06
8	P0.07
9	P0.08
10	P0.09
11	P0.10
12	P0.11
13	P0.12
14	P0.13
15	P0.14
16	P0.15
17	VDD_GPIO
18	VDD_GPIO
19	GND
20	GND

J13 — GPIO_P0.16-31

Pin	Net
1	P0.16
2	P0.17
3	P0.18
4	P0.19
5	P0.20
6	P0.21
7	P0.22
8	P0.23
9	P0.24
10	P0.25
11	P0.26
12	P0.27
13	P0.28
14	P0.29
15	P0.30
16	P0.31
17	VDD_nRF
18	VDD_GPIO
19	GND
20	GND



Sheet: /K. LEDs/BUTTONS/		
File: 11_LEDS_BUTTONS.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 12/13



Sheet: /L. TEST POINTS/		
File: 12_TESTPOINTS.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 13/13